Faculty of Computer Science

Applied Quantitative Logistics

Moscow 2025

A Hybrid GA and PSO Optimized Approach for Heart-Disease Prediction

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Publication: Springer(2022)

Problem Statement:

- Heart disease is one of the leading causes of death worldwide.
- Early detection is crucial to improving outcomes and saving lives.
- This study combines Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Random Forest (RF) to enhance heart disease prediction.

Objective of the Study:

- Select the most significant patient features using GA and PSO.
- Enhance prediction accuracy using Random Forest classifier.
- Reduce computational complexity in feature selection.

Related Works:

Table 1 Summary of the related work

Study	Proposed	Techniques & Tools	Finding	Limitation
Mohammad S. Amin et al. [5]	Developed a heart-disease-prediction model by using the identified best features and data-mining algorithms.	Vote algorithm as a hybrid methodology of logistic regression and NB. (RapidMiner)	They achieved high accuracy in heart-disease prediction on the Cleveland dataset.	The feature selection algorithm is not use.
Saqlain et al. [46]	A cardiac disease diagnosis system is presented by proposing the feature selection algorithms.	Fisher score & Matthews correlation coefficient as an FS algorithm and SVM for binary classification.	The proposed technique achieved considerably better prediction results than the other comparative techniques.	The feature selection algorithm needs to improve.
Purnomo et al. [39]	Predicting heart-disease using different FS approached and NB.	NB + (Backward Elimination / Optimize Selection / Forward Selection).	(NB + forward selection) outperforms other feature selection approach.	The classifier algorithm needs to improve.
Vivekanandan and Iyengar [51]	Optimal feature selection using a modified differential evolution (DE) algorithm and its effectiveness for prediction of heart disease.	Modified differential evolution algorithm and integrated model of fuzzy analytic hierarchy process (AHP) & ANN.	The results show that the modified DE outperforms the best proven traditional DE for feature selection.	The classification accuracy needs to increase.
Priyatharshini and Chitrakala [38]	Build an efficient mining framework for coronary disease diagnosis.	A self-learning approach for a fuzzy rule-based system. (MATLAB)	The experimental results show that the self-learning approach has the efficacy to categorize the risk levels of individuals with suspected coronary disease.	The datasets are limited to validate the experiments.
Halder er al. [21]	A computerized diagnosis system implemented for the classification of myocardial infarction (MI) disease.	Rough set classifier from multi-lead ECG signal.	The proposed method not only finds the suitable rules to explore better knowledge but also the important factors affecting the decision making of MI by using rough set explorer system (RSES).	
Long et al. [30]	A heart-disease-diagnosis system.	A chaos-based firefly algorithm, rough set-based attribute reduction, and an interval type-2 fuzzy logic system for classification. (MATLAB+WEKA)	The proposed combination rough sets-based attribute reduction with interval type-2 fuzzy logic system overcomes others in terms of accuracy, convergence speed and processing time.	The authors need to utilize another hybrid feature selection algorithms and other classifiers to increase accuracy.
Dwivedi [15]	Different machine learning techniques was evaluated for prediction of heart-disease.	ANN, SVM, NB, logistic regression, KNN, and classification trees.	The analysis of heart disease prediction using different machine algorithms.	The feature selection algorithm is not use.
Krishnaiah et al. [29]	Heart-disease prediction system using data mining technique.	An exponential membership function with standard deviation & incorporating fuzzy set methods into the classical KNN decision rule. (WEKA)	The result shows the capable in nature to remove the redundancy of the data and the better accuracy of the system & the performance analysis shows that the fuzzy K-NN classifier	The experiment environment is not mentioned. And the classification accuracy needs to increase.

Related Work Summary

- Many studies did not use proper feature selection, leading to lower accuracy and higher complexity.
- Some approaches used complex classifiers (e.g., deep neural networks) without optimizing input features.
- Overfitting was common due to small dataset sizes or imbalanced classes.
- Very few studies adopted hybrid optimization methods to combine strengths of multiple algorithms.
- Most models ignored rejected solutions that might still contain useful genetic information.
- Lack of cross-validation or reliable performance metrics in some studies limited result generalizability.
- Some methods focused more on classifier design rather than enhancing data quality and input relevance.

Proposed Methodology: GAPSO-RF

Genetic Algorithm and Particle Swarm Optimization with Random Forest

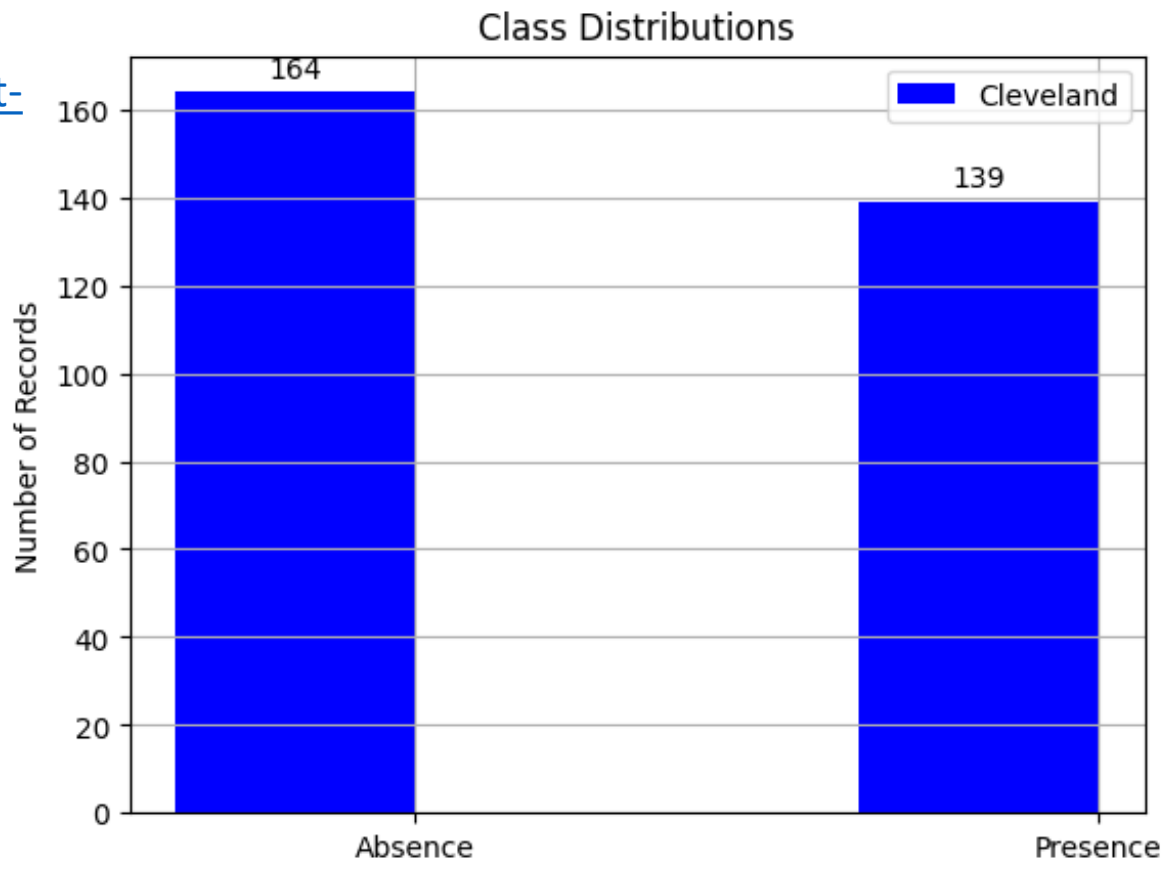
1. Multivariate statistical analysis to rank features.
2. GA selects best features using discriminate mutation.
3. PSO re-evaluates rejected individuals.
4. Random Forest classifies based on selected features.

Dataset:

Cleveland Heart Disease Dataset

Link: https://www.kaggle.com/datasets/ritwikb3/heart-disease-cleveland

- Dataset Source: Sourced from the UCI Machine Learning Repository via Kaggle, contains 303 records and 14 attributes.
- Target Variable: The 'target' indicates the presence (1) or absence (0) of heart disease.
- Preprocessing: Data was normalized using MinMaxScaler; irrelevant columns were excluded to streamline model performance.



Workflow of GAPSO-RF:

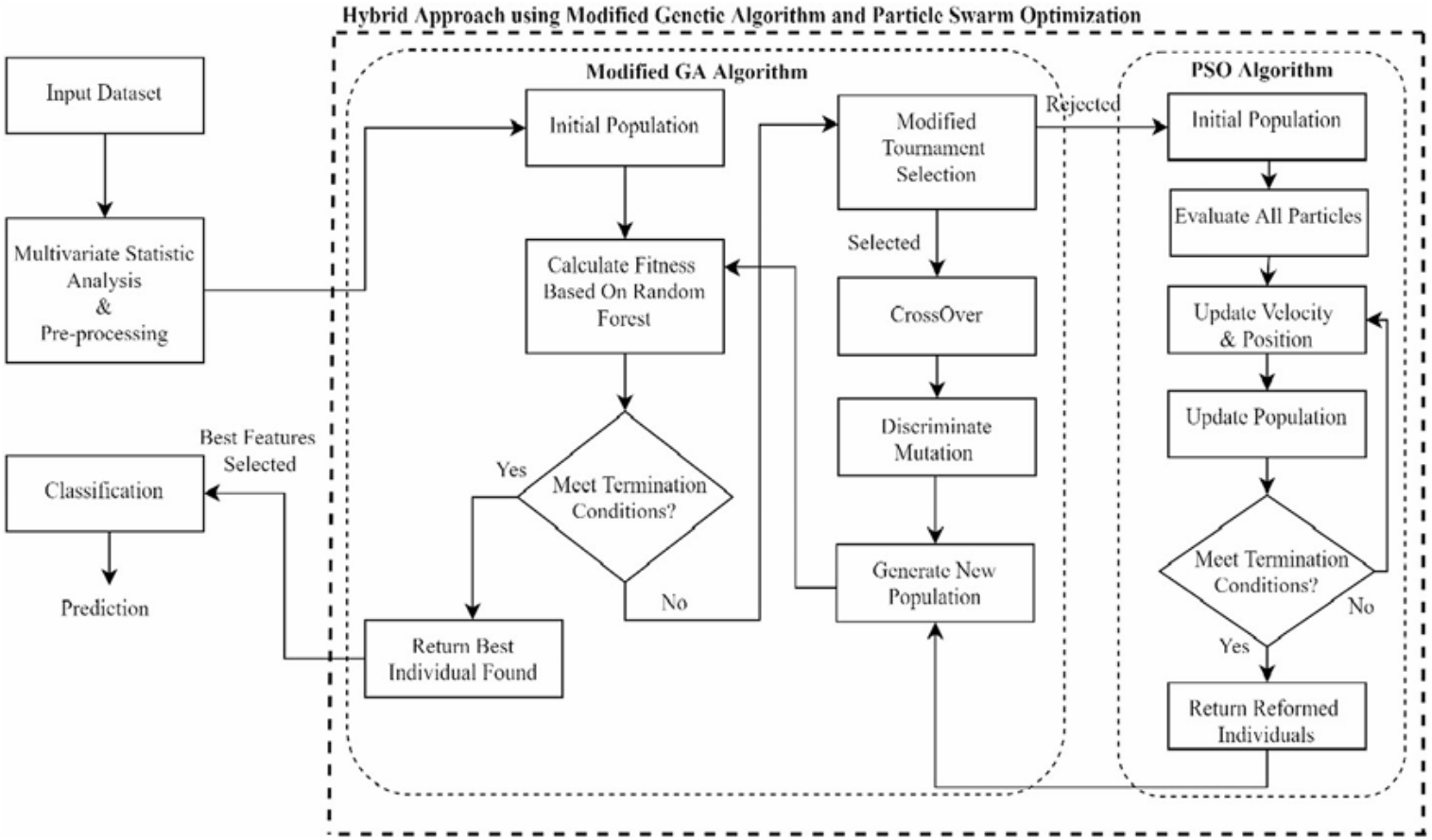


Fig. 1 Workflow of the proposed GAPSO-RF approach

Step-By-Step Implementation:

Data Preprocessing: The data are normalized before performing FS. In the proposed approach, we implemented min–max normalization:

$$a = \frac{a - a_{min}}{a_{max} - a_{min}} \times [max_new - min_new] + min_new;$$

Step-By-Step Implementation:

Modified Genetic Algorithm (GA):

- **Initial Population:** Random solutions (feature subsets) are generated.
- **Fitness Calculation:** Each solution is evaluated using a classification technique.
- **Termination Check:** If the stopping criteria are met (e.g., max generations, accuracy threshold), return the best solution If not, proceed to evolve the population.
- **Modified Tournament Selection:** Chooses promising candidates.
- **Crossover:** Combines selected features from two solutions.
- **Discriminate Mutation:** Intelligently tweaks feature sets to explore better combinations.
- **Generate New Population:** Updated generation is created.
- **Rejected Individuals:** Poor solutions are passed to PSO for further optimization.

Step-By-Step Implementation:

Particle Swarm Optimization (PSO)

- **Initial Population:** Starts with the rejected GA solutions.
- **Evaluate All Particles:** Determines fitness for each feature
- **Update Velocity & Position:** Adjusts each particle (solution) toward the best-found positions.
- **Update Population:** Replaces weaker solutions with better ones.
- **Termination Check:** If met, returns reformed (improved) solutions to GA.

Final Step (Classification):

Random Forest Classifier:

Best features are selected from the optimized solution.

- **Classification:** Final model classifies the input using Random Forest.
- **Prediction:** Output is a heart disease prediction.
- **Metrics used:** Accuracy, Sensitivity, Specificity, AUC

Results & Findings:

Study	Method			Accuracy
	Pre-Processing	Feature Selection	Classification	
El-Bialy et al. [16]	–	Manual	C4.5 algorithm and fast decision tree	78.54 (10-fold)
Paul et al. [35]	handling missing values	Correlation coefficient & GA	fuzzy rules	80 (holdout)
Saqlain et al. [46]	Data standardiza-tion	Fisher score and the Matthews correlation coefficient	SVM	81.19 (10-fold)
Shah et al. [48]	Normalization	parallel probabilistic principal component analysis	SVM	82.18 (10-fold)
Vivekanandan and Iyengar [51]	*min--max* normalization	modified differential evolution algorithm	Integrated model of fuzzy AHP & ANN	83 (holdout)
Suresh and Ananda Raj [50]	handling missing values	GA	NB	83.20 (k-fold)
Buettner et al. [11]	Grouping	–	RF	84.40 (10-fold)
Chitra and Seenivasag-am [12]	*min--max* normalization	–	Feed Forward and Cascaded Correlation NN	85 (holdout)
Mathan et al. [32]	–	–	Gini index-based decision tree & Neural network	85.30 (10-fold)
Jha et al. [23]	–	–	Random subspace classifier	86.14 (3-fold)
Saxena et al. [47]	–	All Possible-MV algo-rithm to handle the missing values	Decision-trees and rules generated	86.70 (10-fold)
Subanya et al. [49]	–	Artificial bee colony algorithm	SVM	86.76 (holdout)
Amin et al. [5]	remove the missing values	Tests 8100 combinations of the features	Vote algorithm as a hybrid methodology of logistic regression and NB	86.87 (10-fold)
Gokulnath and Shantharaj-ah [18]	*Z* score	GA	SVM	88.34 (holdout)
Yazid et al. [54]	–	–	Flower pollination neural network	89.60 (holdout)
Ali et al. [3]	–	–	Hybrid grid search algorithm using two optimized SVM models	92.20 (holdout)
Ali et al. [4]	–	χ^2 statistical model	Deep neural network	93.33 (holdout)
GAPSO-RF (ours)	*min--max* normalization	GA-PSO	RF	**87.80 (10-fold)**
GAPSO-RF (ours)	*min--max* normalization	GA-PSO	RF	**95.60 (holdout)**

Results with Our Implementations:

Multiple Classifiers on the Selected Features

```
Cleveland Classifier Comparison on Selected Features:
       Model  Mean Accuracy  Std Dev
RandomForest       0.844194 0.067596
DecisionTree       0.761505 0.104293
  NaiveBayes       0.808280 0.047495
         SVM       0.818280 0.065861
```

Grid Search for Random Forest hyperparameters

```
RF Best Params & Accuracy by Grid Search
        Parameter     Value
        max_depth         5
     max_features      sqrt
 min_samples_leaf         1
min_samples_split         5
     n_estimators        50
      RF Accuracy  0.861148
```

- Here we see Random Forest with hyperparameters by Grid Search accuracy is better then the Random Forest.

Conclusion:

- Applied on heart disease datasets, the method showed superior performance compared to existing techniques.
- Achieved a highest accuracy of 95.60%, outperforming other models like deep learning and SVM-based hybrids.
- Results confirm that intelligent feature selection and hybrid optimization significantly boost prediction reliability.
- This model can be generalized for other medical prediction problems, making it a promising tool in healthcare analytics.

Planed Future Analysis:

- We'll implement this method on 2nd Dataset(Statlog heart Dataset)
- We'll try to improve accuracy of our implementation
- We'll implement more Algorithms(Simulated Annealing or Harmony Search) and will do comparison analysis

Thank You!